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| **Fiscal Year:** | FY 2007 | **Task Last Updated:** | FY 11/01/2010 |
| **PI Name:** | Karma, Alain Ph.D. | | |
| **Project Title:** | Adaptive Control of Alloy Microstructures | | |
| **Division Name:** | Physical Sciences | | |
| **Program/Discipline:** | MATERIALS SCIENCE | | |
| **Program/Discipline--Element/Subdiscipline:** | MATERIALS SCIENCE--Metals and alloys | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | None | | |
| **Human Research Program Risks:** | None | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **Zip Code:** | 02115-5005 | **Congressional District:** | 8 |
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| **Start Date:** | 11/01/2003 | **End Date:** | 09/30/2007 |
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| **No. of Master's Candidates:** | | **No. of Bachelor's Degrees:** | |
| **No. of Bachelor's Candidates:** | | **Monitoring Center:** | NASA MSFC |
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| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: Received no-cost extension to 9/30/2007 per PI (3/07) | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Losert, Wolfgang ( University of Maryland ) | | |
| **Grant/Contract No.:** | NNM04AA15G | | |
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| **Performance Goal Text:** | | | |
| | Material microstructure is crucial in diverse areas of materials research from metallurgy to studies of biomaterials. In particular, the microstructure (structure on sub-micron to millimeter scales) often determines material strength and other key properties. Our research aims to develop a new approach toward understanding and controlling microstructural patterns. We propose a unique combination of quantitative phase-field modeling and novel perturbation experiments. The project will focus on microstructures of great technological relevance, cellular and dendritic arrays that develop during binary alloy directional solidification, though the basic approach to microstructure control should be more broadly applicable to a wide range of materials. In traditional directional solidification experiments, control of the microstructure is limited to global control through variations of the alloy composition and the growth conditions (externally imposed growth rate and temperature gradient). The evolution of the solid-liquid interface can only be influenced through | | |

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| **Task Description:** | changes of these parameters.<br>We propose to pursue a new control approach where the interface evolution can be altered by spatially localized perturbations that can be applied at arbitrary times. These perturbations are chosen adaptively, i.e. based on the current state of the system, so as to guide the interface evolution into desired microstructures with optimal properties that would not otherwise be accessible. This approach exploits the extreme sensitivity of dendritic patterns to very small amplitude perturbations, of magnitude comparable to thermal fluctuations, and the fact that a wide range of structures can form under the same growth conditions due to the nonequilibrium nature of the solidification process.<br><br>This research relies on new state-of-the-art experimental and computational tools:<br><br>1) A holographic laser tweezer array will be used to apply complex perturbations to directionally solidified impure succinonitrile crystals using a laser light field that can be shaped adaptively in response to the observed microstructure. Real time observations of the microstructure in three-dimensional bulk samples will be made possible by a two-photon confocal microscope, while observations in thin samples will utilize a standard optical microscope set-up. These techniques will allow us to create perturbations of the solid-liquid interface of arbitrary shape and wavelength (i.e. arbitrary spatial arrangements of UV light spots) as well as to locally perturb or remelt a desired subset of cell/dendrite tips in an array.<br><br>2) Simulations of a new alloy phase-field formulation (Karma, 2002b) will be used to model quantitatively the solid-liquid interface evolution in two and three dimensions with realistic materials parameters and to guide the design of the experiments.<br><br>In a past collaboration, the PI and Co-I have already demonstrated that perturbative experiments and phase-field simulations can be combined successfully to obtain new insights into microstructural evolution beyond the beaten path of conventional approaches (Losert et al., 1998c). The use of the aforementioned new techniques will greatly enlarge the scope of this work by extending it to three dimensions and by making the comparison between simulations and experiments quantitative. Our dual objective is to use these tools to shed new light on long standing microstructural pattern formation questions that are of both fundamental and practical importance and to produce new microstructures with desired properties. We will focus our attention specifically on two closely related problems: the dynamical selection of the primary cell/dendrite array spacing, which is influenced by both the complex history of the solid-liquid interface evolution and the sidebranching dynamics, and the cell to dendrite transition in both two and three dimensions. In addition, we will investigate how adaptive control can be used to stabilize unstable microstructures using the planar interface as a first concrete t [task description truncated] |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | |
| **Task Progress:** | In conclusions, we have shown experimentally and numerically that highly unstable cellular solidification arrays can be stabilized using feedback control. Experiments and simulations showed similar dynamical evolution as the array was controlled. Unlike feedback stabilization schemes previously developed for planar fronts slightly above the critical velocity corresponding to the onset of morphological instability, our approach stabilizes efficiently highly unstable states and can be implemented experimentally with only a discrete number of controllable heating points. An alternate control scheme which includes a tunable heating power for each spot has been implemented numerically and the results are presently being written-up for publication. Although a quantitative comparison between numerical and experimental results is not yet possible because of the lack of detailed knowledge of the alloy phase diagram, the observed qualitative behaviors are identical. The use of other localized physical perturbations generated, e.g., by x-rays or ultrasound could potentially make it possible to extend this control scheme to metallic alloys where the growth of much finer array structures with superior mechanical properties is of considerable practical interest. In addition, the ability to access experimentally unstable steady-state patterns provides an important new tool to enrich our fundamental understanding of pattern formation in directional solidification. |
| **Bibliography Type:** | Description: (Last Updated: 05/10/2026) |
| **Articles in Peer-reviewed Journals** | Pons AJ, Karma A, Akamatsu S, Newey M, Pomerance A, Singer H, Losert W. "Feedback control of unstable cellular solidification fronts." Phys Rev E Stat Nonlin Soft Matter Phys. 2007 Feb;75(2 Pt 1):021602. PMID: 17358348 , Feb-2007 |